MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 11 (1-5 Nov 2026) - Period 1	Ch 8 - 8.7 Iteration: pre-condition loops

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 11 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 78% wrote a correct FOR loop with STEP; range() end values weak. Today: WHILE...DO...ENDWHILE when repeats are unknown.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students write pre-condition loops (WHILE ... DO ... ENDWHILE) in pseudocode and Python while, and choose between count-controlled and pre-condition loops. Syllabus 8.1.4(c); AO2, AO3.

Success criteria — *what will successful learning look / sound like?*

I can write a WHILE loop that ends correctly; I can explain why it may run zero times; I can choose FOR or WHILE for a scenario.

Key knowledge / skill / understanding:

Condition tested before each repeat; loop may not run at all; the condition must change inside the loop.

Likely misconception / access or challenge need:

WHILE always runs at least once; infinite loops are syntax errors.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Inquiry / investigation **Time:** 8 min

What students will do:

Predict-then-run: 3 WHILE snippets (normal, never runs, infinite). Predict output, run, explain, learn to stop an infinite loop.

Evidence tool: Mini-whiteboards

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Snippets in Thonny.

Support (low performers / SEN):

Annotated snippets.

Stretch (fast learners / gifted):

Rewrite the infinite loop so it ends.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Problem solving **Time:** 22 min

What students will do:

Program: a savings tracker that adds a monthly amount WHILE Total < Goal and counts months; then a guessing game using WHILE until the guess is right.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Coursebook (CUP 9781108915144) 8.7; Python.

Support (low performers / SEN):

Flowchart of the savings loop.

Stretch (fast learners / gifted):

Add RANDOM target and limit guesses.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Debate **Time:** 7 min

What students will do:

FOR or WHILE? Teams argue for 4 scenarios (reading 30 marks, password entry, game loop, countdown).

Evidence tool: Oral questioning

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Scenario slide.

Support (low performers / SEN):

Sentence starters.

Stretch (fast learners / gifted):

Name a scenario where both work and which is clearer.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: 'Keep stirring WHILE the tea is too hot'. Model WHILE Total < Goal DO ... ENDWHILE and Python while; show how the condition variable changes; show Ctrl+C / Stop for infinite loops.

What is new or innovative in this lesson? *(ETEC looks for this)*

Savings goals use students' own targets (Umrah trip, new phone) to make iteration personal and numeric.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Condition never changes (infinite loop); initialising the variable inside the loop; off-by-one on month count. Likely: 'my program froze'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check: write a WHILE loop that halves a number until it is below 1; explain zero-run case. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Coursebook 8.7 pre-condition questions; program: count how many years to double money at 5% interest.

5 REFLECT — After the lesson *Complete within a day. This is reflective-practice evidence and feeds your next Starting point.*

What worked — and how do I know?

Savings goals were meaningful and varied by background; all classes built the tracker. Predict-then-run caused good surprise.

What will I change next time?

Teach stopping infinite loops before practice; 4 students lost work.

Students needing follow-up:

10BR3: 3 students struggle to initialise counters; small-group support next lesson.

COVERAGE CHECK *Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.*

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☑C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☐H

Create	Discussion	Pair / group	Learner choice	Self-assessment	Peer assessment	SEN / low	Gifted
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

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Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 11 (1-5 Nov 2026) - Period 2	Ch 8 - 8.7 Iteration: post-condition loops

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 11 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit check: 80% wrote a terminating WHILE loop; counter initialisation weak. Today: REPEAT...UNTIL and input validation loops.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students write post-condition loops (REPEAT ... UNTIL) in pseudocode, emulate them in Python (while True / break), and compare all three loop types. Syllabus 8.1.4(c); AO2, AO3.

Success criteria — *what will successful learning look / sound like?*

I can write REPEAT...UNTIL for a validation loop; I can explain it runs at least once; I can code it in Python and compare it with WHILE.

Key knowledge / skill / understanding:

Condition tested after the body; always runs at least once; UNTIL stops when condition is TRUE (opposite of WHILE).

Likely misconception / access or challenge need:

UNTIL condition works like WHILE condition; Python has a REPEAT keyword.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 10 min

What students will do:

Worked example: password entry REPEAT INPUT Password UNTIL Password = "Secret" (syllabus example), then Python while True/break.

Evidence tool: Mini-whiteboards

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Syllabus pseudocode p.45; Python.

Support (low performers / SEN):

Side-by-side pseudocode/Python card.

Stretch (fast learners / gifted):

Add an attempts counter that locks after 3 tries.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Stations / carousel **Time:** 20 min

What students will do:

Three stations (6-7 min): A) range-check validation (mark 0-100); B) menu that repeats until 'Q'; C) convert a WHILE loop to REPEAT and explain the difference.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Station cards; Python.

Support (low performers / SEN):

Station cards with code skeletons.

Stretch (fast learners / gifted):

At C, prove with a test where WHILE and REPEAT give different results.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Cooperative learning **Time:** 7 min

What students will do:

Groups complete a comparison table: FOR vs WHILE vs REPEAT (when tested, minimum runs, best use, example).

Evidence tool: Peer-assessment checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Comparison table template.

Support (low performers / SEN):

Table with examples to sort.

Stretch (fast learners / gifted):

Add Python equivalent for each.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: an ATM asks for PIN at least once. Model REPEAT ... UNTIL with the syllabus layout and the inverted condition; then Python while True: ... if condition: break. Explain station rotation.

What is new or innovative in this lesson? *(ETEC looks for this)*

Validation stations connect Chapter 7 validation algorithms to real code; students build a class comparison table used for revision.

Progress recorded / shared:

☑ Gradebook ☑ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Inverted condition errors (UNTIL vs WHILE logic); break inside wrong block; validation not re-prompting. Likely: 'my loop stops immediately'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: write REPEAT...UNTIL for an age 11-18 range check; state minimum runs for each loop type. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Coursebook 8.7 post-condition questions; complete loop comparison table in notebook.

5 REFLECT — After the lesson *Complete within a day. This is reflective-practice evidence and feeds your next Starting point.*

What worked — and how do I know?

Stations kept pace high; comparison tables were strong and shared on Teams. 75% completed all three stations.

What will I change next time?

Teach the UNTIL condition inversion with a visual (stop sign) before stations.

Students needing follow-up:

Students who found station C hard (approx. 3 per class) get a WHILE-REPEAT conversion sheet.

COVERAGE CHECK *Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.*

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☐H
Create	**Discussion**	**Pair / group**	**Learner choice**	**Self-assessment**	**Peer assessment**	**SEN / low**	**Gifted**
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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